Exercise - 13.1

3. Show that the following points are collinear.

a) (3, -2, 4), (1, 1, 1) and (-1, 4, -2)

⇒ Soln,

Given points are A(3, -2, 4), B(1, 1, 1) and C(-1, 4, -2)

Using distance Formula,

$$|AB| = \sqrt{(1-3)^2 + (1+2)^2 + (1-4)^2}$$
$$= \sqrt{(-2)^2 + 3^2 + (-3)^2}$$
$$= \sqrt{4+9+9}$$
$$= \sqrt{22} \text{ units.}$$

$$|BC| = \sqrt{(-1-1)^2 + (4-1)^2 + (-2-1)^2}$$
$$= \sqrt{(-2)^2 + 3^2 + (-3)^2}$$
$$= \sqrt{4+9+9}$$
$$= \sqrt{22} \text{ units}$$

$$|AC| = \sqrt{(-1-3)^2 + (4+2)^2 + (-2-4)^2}$$
$$= \sqrt{4^2 + 6^2 + (-6)^2}$$
$$= \sqrt{16+36+36}$$
$$= \sqrt{88}$$
$$= 2\sqrt{22} \text{ units}$$

Here, $|AB| + |BC| = |AC|$. Hence, the given three points are collinear.

b) $(1,-2,3)$, $(2,3,-4)$ and $(0,+7,-10)$

→ Soln,

Given points are $A(1,-2,3)$, $B(2,3,-4)$ and $C(0,-7,-10)$

Using distance Formula,

$$|AB| = \sqrt{(2-1)^2 + (3+2)^2 + (-4-3)^2}$$
$$= \sqrt{1^2 + 5^2 + (-7)^2}$$
$$= \sqrt{1+25+49}$$
$$= \sqrt{75} \text{ units.}$$

$$|BC| = \sqrt{(0-2)^2 + (-7-3)^2 + (+10+4)^2}$$
$$= \sqrt{(-2)^2 + (-10)^2 + \cancel{(6)^2}\ 14^2}$$
$$= \sqrt{4 + 100 + 196}$$
$$= \sqrt{300} \text{ units}$$

$$AC = \sqrt{(0-1)^2 + (-7+2)^2 + (+10-3)^2}$$
$$= \sqrt{(-1)^2 + (-5)^2 + (7)^2}$$
$$= \sqrt{1 + 25 + 49}$$
$$= \sqrt{75} \text{ units}$$

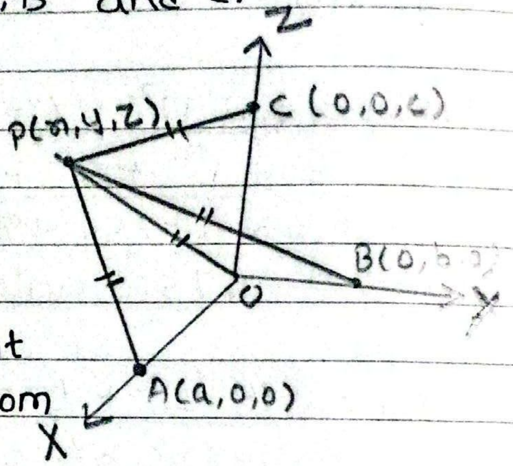
$$|AB| + |AC| = |BC|$$
$$\sqrt{75} + \sqrt{75} = \sqrt{300}$$
$$2\sqrt{75} = 10\sqrt{3}$$
$$10\sqrt{3} = 10\sqrt{3}$$

Here, $|AB| + |AC| = |BC|$. Hence, the given three points are collinear.

4) If A, B and C are the points on X, Y and Z axis respectively at distance a, b and c from the origin O. Find the co-ordinates of a point P which is equidistant from the four points O, A, B and C.

→ Soln,

Here, coordinates of O(0,0,0), A(a,0,0), B(0,b,0) and C(0,0,c)




let P(x,y,z) be a point which is equidistant from the four points O, A, B and C.

i.e. $|OP| = |AP| = |BP| = |CP|$

Now, taking $|OP| = |AP|$

$$\text{or, } \sqrt{x^2+y^2+z^2} = \sqrt{(x-a)^2+y^2+z^2}$$

~~or,~~ Squaring on both sides,

$$x^2+\cancel{y^2}+\cancel{z^2} = (x-a)^2+\cancel{y^2}+\cancel{z^2}$$

$$\text{or, } \cancel{x^2} = \cancel{x^2} - 2ax + a^2$$

$$\text{or, } a^2 = 2ax$$

$$\text{or, } x = \frac{a^{\cancel{2}}}{2\cancel{a}}$$

$$\therefore x = \frac{a}{2}$$

Again, taking $|OP| = |BP|$

or, $\sqrt{x^2+y^2+z^2} = \sqrt{x^2+(y-b)^2+z^2}$

Squaring on both sides,

$\cancel{x^2}+y^2+\cancel{z^2} = \cancel{x^2}+(y-b)^2+\cancel{z^2}$

or, $\cancel{y^2} = \cancel{y^2} - 2by + b^2$

or, $b^2 = 2by$

or, $y = \frac{b^{\cancel{2}}}{2\cancel{b}}$

$\therefore y = \frac{b}{2}$

Taking $|OP| = |CP|$

or, $\sqrt{x^2+y^2+z^2} = \sqrt{x^2+y^2+(z-c)^2}$

Squaring on both sides,

$\cancel{x^2}+\cancel{y^2}+z^2 = \cancel{x^2}+\cancel{y^2}+(z-c)^2$

or, $\cancel{z^2} = \cancel{z^2} - 2cz + c^2$

or, $2\cancel{c}z = c^{\cancel{2}}$

or, $z = \frac{c}{2}$

$\therefore$ Coordinates of P are $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ //.

5.

c) If the internal section of two points $P(2,-4,3)$ and $Q(x,y,z)$ in the ratio $2:1$ is $(-2,2,-3)$, find $Q$.

→ Soln,

Given,

$P(2,-4,3) = (x_1, y_1, z_1)$

$Q(x,y,z) = (x_2, y_2, z_2)$

$m_1 : m_2 = 2:1$

coordinates of dividing point $= (-2,2,-3)$

Using section formula,

Coordinates of dividing point =

$$\left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

or, $(-2,2,-3) = \left(\frac{2x+2}{2+1}, \frac{2y-4}{2+1}, \frac{2z+3}{2+1}\right)$

or, $(-2,2,-3) = \left(\frac{2x+2}{3}, \frac{2y-4}{3}, \frac{2z+3}{3}\right)$

Equating the corresponding components, we get

$-2 = \frac{2x+2}{3}$

or, $-6 = 2x+2$

or, $2x = -6-2$

or, $x = \frac{-8}{2}$

$\therefore x = -4$

$2 = \frac{2y-4}{3}$

or, $6 = 2y - 4$

or, $2y = 10$

or, $y = \frac{10}{2}$

$\therefore y = 5$

Now,

$-3 = \frac{2z+3}{3}$

or, $-9 = 2z+3$

or, $2z = -12$

or, $z = -6$

$\therefore z = -6$

$\therefore$ The coordinates of point Q are $(-4, 5, -6)$ //.

1. Find the co-ordinates of the mid points of the line joining the points.

a) $(-2, 6, -4)$ and $(4, 0, 8)$

→ Soln,

$$\text{Midpoint } (x,y,z) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

$$= \left(\frac{-2+4}{2}, \frac{6+0}{2}, \frac{-4+8}{2}\right)$$

$$= \left(\frac{2}{2}, \frac{6}{2}, \frac{4}{2}\right)$$

$$= (1, 3, 2)$$

$\therefore$ Midpoint $(x,y,z) = (1,3,2)$ //.

b) $(-1, -2, -1)$ and $(4, 7, 6)$

→ Soln,

$$\text{Midpoint } (x,y,z) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

$$= \left(\frac{-1+4}{2}, \frac{-2+7}{2}, \frac{-1+6}{2}\right)$$

$$= \left(\frac{3}{2}, \frac{5}{2}, \frac{5}{2}\right)$$

$\therefore$ Midpoint $(x,y,z) = \left(\frac{3}{2}, \frac{5}{2}, \frac{5}{2}\right)$ //

2. Find the distance between the points:

a) $(-2,1,0)$ and $(3,5,-2)$

→ let, $A = (-2,1,0)$ and $B = (3,5,-2)$

$$|AB| = \sqrt{(3+2)^2 + (5-1)^2 + (-2-0)^2}$$

$$= \sqrt{5^2 + 4^2 + (-2)^2}$$

$$= \sqrt{25+16+4}$$

$$= \sqrt{45}$$

$$= 3\sqrt{5} \text{ //}$$

b) $(-4,7,-7)$ and $(-2,1,-10)$

→ let $A = (-4,7,-7)$ and $B = (-2,1,-10)$

$$|AB| = \sqrt{(-2+4)^2 + (1-7)^2 + (-10+7)^2}$$

$$= \sqrt{(-2)^2 + (-6)^2 + (-3)^2}$$

$$= \sqrt{4+36+9}$$

$$= \sqrt{49}$$

$$= 7 \text{ units //}$$

Qn. Find the co-ordinates of a point where the line joining the points A(2, -4, 3) and B(4, 2, -3). Find the co-ordinates of point where the line joining A(2, -4, 3) and B(4, 2, -3) cuts the XZ-plane.

→ Solution,

Given,

$A(2,-4,3) = (x_1, y_1, z_1)$

$B(4,2,-3) = (x_2, y_2, z_2)$

let $P(x, 0, z)$ be a point on XZ-plane Which divides AB in the ratio of $m_1 : m_2$.

so,

y-coordinates of dividing point

$$= \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

or, $0 = \dfrac{m_1 \times 2 + m_2 \times (-4)}{m_1 + m_2}$

or, $0 = 2m_1 - 4m_2$

or, $4m_2 = 2m_1$

or, $\dfrac{m_1}{m_2} = \dfrac{4}{2}$

$\therefore m_1 : m_2 = 2 : 1$

Now, the coordinates of dividing point P are

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}, \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2}\right)$$

$$= \left(\frac{2\times 4 + 1\times 2}{2+1}, \frac{2\times 2 + 1\times(-4)}{2+1}, \frac{2\times(-3) + 1\times 3}{2+1}\right)$$

$$= \left(\frac{8+2}{3}, 0, \frac{-6+3}{3}\right)$$

$$=\left(\frac{10}{3}, 0, -1\right) \#.$$

6.

d) Find the point where the line joining the points $(2,-3,1)$ and $(3,-4,-5)$ cuts the plane $2x+y+z=7$.

→ Solution

Given,

$A(2,-3,1) = (x_1, y_1, z_1)$

$B(3,-4,-5) = (x_2, y_2, z_2)$

And the equation of plane is $2x+y+z=7$ — (i)

Let $P(x,y,z)$ be a point on plane (i) which divides AB in the ratio of $m_1 : m_2$. So,

$$(x,y,z) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

$$(x,y,z) = \left(\frac{3m_1+2m_2}{m_1+m_2}, \frac{-4m_1-3m_2}{m_1+m_2}, \frac{-5Z_2+Z_1}{m_1+m_2}\right)$$

this point lies on plane (i) so, it must satisfy the equation of plane.

i.e.: $2\left(\frac{3m_1+2m_2}{m_1+m_2}\right)+\left(\frac{-4m_1-3m_2}{m_1+m_2}\right)+\frac{-5m_1+m_2}{m_1+m_2}=7$

or, $\frac{6m_1+4m_2-4m_1-3m_2-5m_1+m_2}{m_1+m_2}=7$

or, $-3m_1+2m_2 = 7m_1+7m_2$

or, $10m_1 = -5m_2$

or, $\frac{m_1}{m_2} = \frac{-5}{10}$

$\therefore m_1 : m_2 = 1 : -2$ //.

Now, the coordinates of dividing point P becomes

$$\left(\frac{3\times1+2\times(-2)}{1-2}, \frac{-4\times1-3\times(-2)}{1-2}, \frac{-5\times1+(-2)}{1-2}\right)$$

$$= \left(\frac{-1}{-1}, \frac{-4+6}{-1}, \frac{-7}{-1}\right)$$

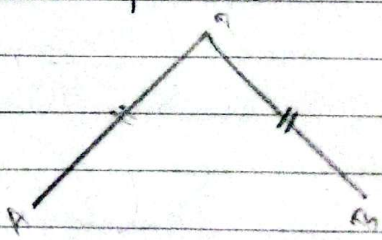
$$= (1, -2, 7)$$ //.

7.

b) ~~Solution~~, Find the Locus of a point which is equidistance from two fixed points (1,2,1) and (3,-4,2)

→ Given points are A(1, 2, 1) and B(3,-4,2). Let (x,y,z) be a moving point equidistant from the given two points.

~~ie~~ i.e: $|AP| = |BP|$




or, $\sqrt{(x-1)^2+(y-2)^2+(z-1)^2} = \sqrt{(x-3)^2+(y+4)^2+(z-2)^2}$

~~or~~ Squaring on both sides, we get

$(x-1)^2+(y-2)^2+(z-1)^2 = (x-3)^2+(y+4)^2+(z-2)^2$

or, $\cancel{x^2}-2x+1+\cancel{y^2}-4y+\cancel{4}+\cancel{z^2}-2z+1 = \cancel{x^2}-6x+9+\cancel{y^2}$

$+8y+16+\cancel{z^2}-4z+\cancel{4}$

or, $-2x-4y-2z+2=-6x+8y-4z+25$

$\therefore 4x-12y+2z=23$ is the required locus.

5.

a) Find the co-ordinate of the point which divides the join of the points (3,3,1) and (3,-6,4) internally in the ratio 2:1.

→ Soln,

let, $A(3,3,1)=(x_1,y_1,z_1)$

$B(3,-6,4)=(x_2,y_2,z_2)$

$m_1:m_2=2:1$

Co-ordinates of dividing point $(x,y,z)=?$

We know that,

$$(x,y,z)=\left(\frac{m_1x_2+m_2x_1}{m_1+m_2},\frac{m_1y_2+m_2y_1}{m_1+m_2},\frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

$$=\left(\frac{3\times2+1\times3}{2+1},\frac{2\times-6+1\times3}{2+1},\frac{2\times4+1\times1}{2+1}\right)$$

$$=\left(\frac{6+3}{3},\frac{-12+3}{3},\frac{8+1}{3}\right)$$

$$=(3,-3,3)$$

$\therefore$ Co-ordinates of dividing point are (3,-3,3) #.

b) Find the co-ordinates of the point which divides the join of the points $(3,4,-5)$ and $(1,3,-2)$ externally in the ratio $5:4$.

→ Soln,

let, $A(3,4,-5) = (x_1, y_1, z_1)$

$B(1,3,-2) = (x_2, y_2, z_2)$

$m_1 : m_2 = 5:4$

co-ordinates of dividing ~~part~~ point $(x,y,z)$ $=?$

We know that,

$$(x,y,z) = \left( \frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 + m_2y_1}{m_1 - m_2}, \frac{m_1z_2 + m_2z_1}{m_1 - m_2} \right)$$

$$= \left( \frac{5\times1 - 4\times3}{5-4}, \frac{5\times3 - 4\times4}{5-4}, \frac{5\times-2 - 4\times-5}{5-4} \right)$$

$$= \left( \frac{5-12}{1}, \frac{15-16}{1}, \frac{-10+20}{1} \right)$$

$$= (-7, -1, 10)$$

$\therefore$ co-ordinates of dividing point are

$(-7, -1, 10)$ #.

6.

a) Given three collinear points $A(3,2,-4)$, $B(5,4,-6)$ and $C(9,8,-10)$. Find the ratio in which B divides AC.

→ Soln,

let $A(3,2,-4) = (x_1, y_1, z_1)$

$C(9,8,-10) = (x_2, y_2, z_2)$

Here, B divides AC so,

$B(5,4,-6) = (x, y, z)$

$m_1 : m_2 = ?$

We know that,

$$(x,y,z) = \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

$$(5,4,-6) = \left(\frac{m_1.9+m_2.3}{m_1+m_2}, \frac{m_1.8+m_2.2}{m_1+m_2}, \frac{m_1.-10+m_2.-4}{m_1+m_2}\right)$$

Equating corresponding component,

$$5 = \frac{9m_1+3m_2}{m_1+m_2}$$

or, $5m_1+5m_2 = 9m_1+3m_2$

or, $4m_1 = 2m_2$

or, $\frac{m_1}{m_2} = \frac{1}{2}$

∴ The ratio in which B divides AC is 1 : 2

b) Find the ratio in which the $xy$-plane divides the join of the points $(-2,4,7)$ and $(3,-5,-8)$

→ Soln,

let, $A(-2,4,7) = (x_1, y_1, z_1)$

$B(3,-5,-8) = (x_2, y_2, z_2)$

XY plane divides so the dividing point are $P(x, y, 0)$

Now,

$$z = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2}$$

or, $$0 = \frac{m_1 \cdot -8 + m_2 \cdot 7}{m_1 + m_2}$$

or, $0 = -8m_1 + 7m_2$

or, $8m_1 = 7m_2$

or, $\frac{m_1}{m_2} = \frac{7}{8}$

$\therefore$ Ratio is $7:8$ #

c) Find the co-ordinates of a point where the line through the point $A(1,2,3)$ and $B(4,-4,9)$ meets the $zx$-plane.

→ Soln,

$A(1,2,3) = (x_1, y_1, z_1)$

$B(4,-4,9) = (x_2, y_2, z_2)$

ZX plane divides so the dividing point are $P(x, 0, z)$

Now,

$$y = 0 = \frac{m_1 . y_2 + m_2 . y_1}{m_1 + m_2}$$

$$\text{or, } 0 = \frac{m_1 . -4 + m_2 . 2}{m_1 + m_2}$$

$$\text{or, } 0 = -4m_1 + 2m_2$$

$$\text{or, } 4m_1 = 2m_2$$

$$\text{or, } \frac{m_1}{m_2} = \frac{1}{2}$$

$$\therefore m_1 : m_2 = 1:2$$

Again,

$$(x, 0, z) = \left( \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, 0, \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \right)$$

$$= \left( \frac{1.4 + 2.1}{1+2}, 0, \frac{1.9 + 2 \times 3}{1+2} \right)$$

$$= \left( \frac{4+2}{3}, 0, \frac{9+6}{3} \right)$$

$$= \left( \frac{6}{3}, 0, \frac{15}{3} \right)$$

$$= (2, 0, 5)$$

$\therefore$ Co-ordinates where zx-plane meets is $(2, 0, 5)$

~~d) Find the point where the Line joining the points (2, -3, 1) and (3, -4, -5) cuts the plane~~

7.

a) Find the Locus of a point which moves such that its distance from the fixed point (2,-1,3) is always 3 units.

→ Soln,

Fixed point (A) = (2,-1,3)

distance (d) = 3 units

moving point (B) = (x,y,z)

Now,

$d = |AB|$

or, $3 = \sqrt{(x-2)^2 + (y+1)^2 + (z-3)^2}$

or, $3 = \sqrt{x^2 - 4x + 4 + y^2 + 2y + 1 + z^2 - 6z + 9}$

Squaring on both sides,

$\cancel{9} = x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 + \cancel{9}$

or, $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0$

$\therefore x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0$ is the required Locus #.

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c) Find the Locus of point $P(x,y,z)$ satisfying the condition $PA^2 + PB^2 = 6$ where $A(-1,2,-1)$ and $B(0,3,-2)$ are two fixed points.

→ Soln,

$A = (-1,2,-1)$

$B = (0,3,-2)$

$P = (x,y,z)$

$|PA| = \sqrt{(x+1)^2 + (y-2)^2 + (z+1)^2}$

$|PA| = \sqrt{x^2+2x+1 + y^2-4y+4 + z^2+2z+1}$

$PA^2 = x^2+y^2+z^2+2x-4y+2z+6$

Similarly,

$|PB| = \sqrt{(x-0)^2 + (y-3)^2 + (z+2)^2}$

$= \sqrt{x^2+y^2-6y+9+z^2+4z+4}$

$PB^2 = x^2+y^2+z^2-6y+4z+13$

Now,

$PA^2+PB^2 = 6$

or, $x^2+y^2+z^2+2x-4y+2z+\cancel{6}+x^2+y^2+z^2-6y+4z+13 = \cancel{6}$

or, $2x^2+2y^2+2z^2+2x-10y+6z+13=0$

∴ The required Locus of point P satisfying the condition $PA^2+PB^2=6$ is

$2x^2+2y^2+2z^2+2x-10y+6z+13=0$.

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8) Prove that the points $(-4,9,6)$, $(0,7,10)$ and $(-1,6,6)$ are the vertices of a right angled isosceles triangle.

→ Soln,

Let, $A = (-4,9,6)$
$B = (0,7,10)$
$C = (-1,6,6)$

$$|AB| = \sqrt{(0+4)^2 + (7-9)^2 + (10-6)^2}$$
$$= \sqrt{4^2 + (-2)^2 + 4^2}$$
$$= \sqrt{16+4+16}$$
$$= \sqrt{36}$$
$$= 6 \text{ unit}$$

$$|BC| = \sqrt{(-1-0)^2 + (6-7)^2 + (6-10)^2}$$
$$= \sqrt{(-1)^2 + (-1)^2 + (-4)^2}$$
$$= \sqrt{1+1+16}$$
$$= \sqrt{18}$$

$$|AC| = \sqrt{(-1+4)^2 + (6-9)^2 + (6-6)^2}$$
$$= \sqrt{3^2 + (-3)^2 + 0^2}$$
$$= \sqrt{9+9}$$
$$= \sqrt{18}$$

Now,
$$|AB|^2 = |BC|^2 + |AC|^2$$
or, $6^2 = (\sqrt{18})^2 + (\sqrt{18})^2$
or, $36 = 18 + 18$
or, $36 = 36$

Since, $|AC| = |BC| = \sqrt{18}$ it is isosceles triangle and it follow pythagoras theorem so, it it Right angled isosceles triangle.

9.

a) Show that the points (1,2,3), (−1,−2,−1), (2,3,2) and (4,7,6) are the vertices of a parallelogram. Also show that this is not a rectangle.

→ Soln,

Let, A(1,2,3), B(−1,−2,−1), C(2,3,2) and D(4,7,6) are the vertices of a parallelogram.

Mid point of AC diagonal

$$(x,y,z) = \left(\frac{1+2}{2}, \frac{2+3}{2}, \frac{3+2}{2}\right)$$

$$= \left(\frac{3}{2}, \frac{5}{2}, \frac{5}{2}\right)$$

Mid point of BD diagonal

$$(x',y',z') = \left(\frac{-1+4}{2}, \frac{-2+7}{2}, \frac{-1+6}{2}\right)$$

$$= \left(\frac{3}{2}, \frac{5}{2}, \frac{5}{2}\right)$$

Since, Mid point of two different diagonal is same so, ABCD are the vertices of a parallelogram.

Again,

$$|AC| = \sqrt{(2-1)^2 + (3-2)^2 + (2-3)^2}$$

$$= \sqrt{1^2 + 1^2 + (-1)^2}$$

$$= \sqrt{1+1+1}$$

$$= \sqrt{3} \text{ unit}$$

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$$|BD| = \sqrt{(4+1)^2 + (7+2)^2 + (6+1)^2}$$

$$= \sqrt{5^2 + 9^2 + 7^2}$$

$$= \sqrt{25 + 81 + 49}$$

$$= \sqrt{155}$$

Here, diagonal AC and diagonal BD have different unit, so we can say this is not a rectangle.

b) Three consecutive vertices of a parallelogram ABCD are A(-5,5,2), B(-9,-1,2) and C(-3,-3,0). Find the co-ordinates of the fourth vertex.

→ Soln,

let, A(-5,5,2), B(-9,-1,2), C(-3,-3,0) and D(x,y,z) be the vertices of a parallelogram.

Now,

Mid point of AC diagonal

$$(x', y', z') = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

$$= \left(\frac{-5-3}{2}, \frac{5-3}{2}, \frac{2+0}{2}\right)$$

$$= (-4, 1, 1)$$

∴ Mid point of AC is (-4,1,1)

Mid point of two diagonal of parallelogram ~~of~~ is same so, mid point of BD diagonal is also (-4,1,1)

Here,

$$(-4,1,1) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

$$= \left(\frac{-9+x}{2}, \frac{-1+y}{2}, \frac{2+z}{2}\right)$$

Calculating the consecutive values, we get

| | | |
|---|---|---|
| $-4 = \frac{-9+x}{2}$ | $1 = \frac{-1+y}{2}$ | $1 = \frac{2+z}{2}$ |
| or, $-8 = -9+x$ | $2 = -1+y$ | $2 = 2+z$ |
| $\therefore x = 1$ | $\therefore y = 3$ | or, $z = 0$ |

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$\therefore$ The co-ordinates of the fourth vertex are $(1,3,0)$ #.

10) Find the third Vertex of the triangle whose centroid is $(7,-2,5)$ and other two vertices are $(2,6,-4)$ and $(15,-10,16)$

$\rightarrow$ Soln,

let, $A(2,6,-4)$, $B(15,-10,16)$ and $C(x,y,z)$ are vertex of the triangle and $G(7,-2,5)$ ~~at~~ is the centroid of a triangle,

Now,

$$G(7,-2,5) = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$$

$$(7,-2,5) = \left(\frac{2+15+x}{3}, \frac{6-10+y}{3}, \frac{-4+16+z}{3}\right)$$

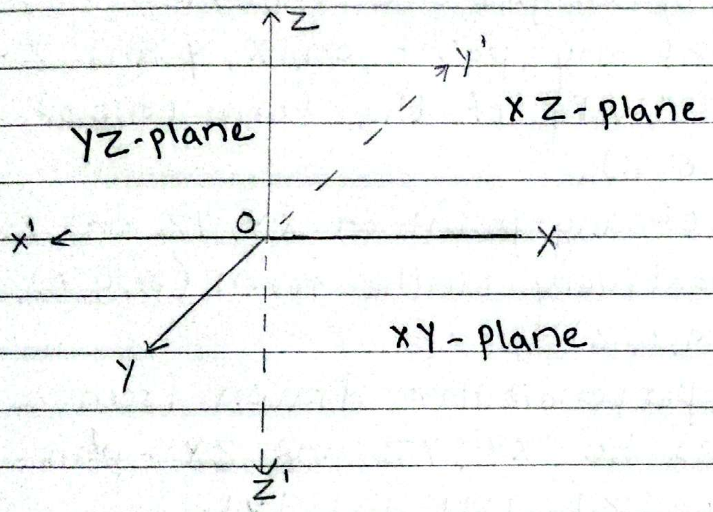
Calculating the consecutive values, we get.

| | | |
|---|---|---|
| $7 = \frac{17+x}{3}$ | $-2 = \frac{-4+y}{3}$ | $5 = \frac{12+z}{3}$ |
| $21-17 = x$ | $-6 = -4+y$ | $15 = 12+z$ |
| $\therefore x = 4$ | $\therefore y = -2$ | $\therefore z = 3$ |

$\therefore$ Third vertex of a triangle is $(4,-2,3)$ #.

# Chapter-13 Co-ordinates in Space

Let XOX', YOY' and ZOZ' be three mutually perpendicular straight lines intersecting at point 'O' respectively called X-axis, y-axis and Z-axis. The point 'O' is called origin and its coordinates are (0,0,0).




The plane formed by X-axis and y-axis is called XY-plane, Y-axis and Z-axis is called YZ-plane and Z-axis and x-axis is called xz-plane. These three coordinate axes divides the space into eight regions called octants. The sign of coordinates in these octants are as follows:

| Coordinate/ octants | I | II | III | IV | V | VI | VII | VIII |
|---|---|---|---|---|---|---|---|---|
| x | + | − | − | + | + | − | − | + |
| y | + | + | − | − | + | + | − | − |
| z | + | + | + | + | − | − | − | − |

The coordinates of a point on space is determined by drawing perpendiculars from that point to YZ-plane, XZ-plane and XY plane. These perpendiculars respectively gives the $x, y$. and Z-coordinates of point.

## Notes

- The coordinates of origin are $(0,0,0)$.
- The coordinates of any point on X, Y and Z-axes are respectively of the form $(x,0,0)$, $(0,y,0)$ and $(0,0,z)$.
- The coordinates of any point on XY, YZ and ZX planes are respectively of the form $(x,y,0)$, $(0,y,z)$ and $(x,0,z)$.
- The length of perpendicular drawn from a point $P(x,y,z)$ to XY, YZ and ZX-planes are respectively $|z|$, $|x|$ and $|y|$.
- The length of perpendicular drawn from a point $P(x,y,z)$ to X, Y and Z-axes are respectively $\sqrt{y^2+z^2}$, $\sqrt{x^2+z^2}$ and $\sqrt{x^2+y^2}$

# List of Formulae:

① distance Formula

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

② Mid point formula

$$(x,y,z) = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

## ③ Section Formula

Internally $= \left( \dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}, \dfrac{m_1z_2+m_2z_1}{m_1+m_2} \right)$

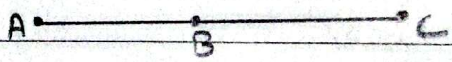
Externally $= \left( \dfrac{m_1x_2-m_2x_1}{m_1+m_2}, \dfrac{m_1y_2-m_2y_1}{m_1+m_2}, \dfrac{m_1z_2-m_2z_1}{m_1+m_2} \right)$

## ④ Centroid

$C = \left( \dfrac{x_1+x_2+x_3}{3}, \dfrac{y_1+y_2+y_3}{3}, \dfrac{z_1+z_2+z_3}{3} \right)$

## * Collinear Points

↳ A set of points lying on a same straight line are called collinear points.




In the figure, the three points A, B and C are collinear. So, $|AC| + |BC| = |AB|$